Autobuild Virtualization Platform

Executive Presentation

- Objective
  - To deliver a full-featured, open-source virtualization platform that includes:
    - detailed node management,
    - automated provisioning,
    - integrated snapshots
    - and comprehensive backup/restore capabilities.

Title Slide

- Autobuild: The Future of Open-Source Virtualization
- Modernizing Infrastructure Through Automation and Open Standards
  - Objective: Full-featured virtualization with detailed node management, snapshots, and backup/restore.
  - Platform: Makefile-driven KVM/Libvirt Ecosystem
  - Presenter: Thomas Sandholm
  - Date: July 7, 2026

Stategic Vision

- A Full-Featured Virtualization Ecosystem
- Our Goal: To bridge the gap between complex enterprise virtualization and the simplicity of open-source toolkits.
  - Open Source Sovereignty: Built on industry-standard KVM and Libvirt to ensure no vendor lock-in.
  - Feature Richness: Delivering enterprise-grade features (Snapshotting, Backups, Role-based Config) in a lean, manageable package.
  - Operational Excellence: Reducing manual intervention through aggressive “Infrastructure as Code” (IaC) principles.
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Challenges in Modern Virtualization

- Complexity and Manual Overhead
- Traditional virtualization management often suffers from:
  - Provisioning Bottlenecks: Manual installation and configuration are slow and error-prone.
  - Configuration Drift: Inconsistent node setups leading to “snowflake” servers.
  - Management Silos: Difficulty in managing snapshots, backups, and network configurations across disparate nodes.
  - Scaling Friction: High effort required to deploy multiple nodes with specific roles (Docker, Jenkins, Database).

Solution Architecture

- Autobuild Platform: Power and Simplicity
- A Makefile-driven orchestration layer that sits atop the industry's most reliable virtualization stack.
  - Hypervisor: KVM (Kernel-based Virtual Machine) for near-native performance.
  - Management Engine: Libvirt for robust API-driven control.
  - Orchestration: GNU Makefile logic for predictable, repeatable, and fast deployments.
  - Guest OS: Automated Ubuntu Cloud Image integration for modern, cloud-native workloads.

Automated Provisioning

- Role-Based Deployment at Scale
- Transforming raw cloud images into production-ready nodes in seconds.
  - Role Driven Customization: Define a ROLE (e.g., docker, jenkins, gluster) and the platform automatically injects the correct packages, boot commands,and mounts.
  - Cloud-Init Integration: Seamlessly stitches together user-data, meta-data and network-config.
  - Backing Images: Uses QCOW2 overlays on shared base images to minimize disk usage and speed up cloning.

Advanced Storage Management

- Tailored Node Architectures
- Detailed disk management beyond just a root filesystem.
  - Multi-Disk Provisioning: Automatically creates and attaches specialed disks:
    - Root Disk: Base OS with flexible resizing.
    - Swap Disk: Optimized memory management.
    - Data Disk: Persistent storage for user data (/home).
    - Database/Web Disks: Dedicated volumes for high-performance I/O applications.
  - Automated Allocation: Disk sizes are configurable via simple Makefile Variables.

Enterprise Reliability

- Integrated Snapshots & Backup/Restore
- Ensuring business continuity with built-in recovery tools.
  - Automated Snapshotting: A “fresh” snapshot is created immediately upon first boot, allowing for instant “reset to factory.”
  - Snapshot Management: Simple commands to create auto-snap snapshots during development or maintenance.
  - Comprehensive Backup: Full backup of VM metadata (XML) and disk images into compressed archives.
  - One-Command Restore: Recover a lost or corrupted node with a single make restore command.

Detailed Node Management

- Network Flexibility & Ansible Integration
- Managing nodes is more than just running them; it’s about integrating them.
  - Network Agility: Support for both Static (mapped via DNS/Hosts) and DHCP configurations.
  - Ansible Ecosystem: Automatic registration of new nodes into Ansible inventories.
  - Inventory Sync: Seamlessly pushes host configurations to central management servers.
  - SSH Lifecycle: Integrated sshresesest and key injections for secure, immediate access.

User-Centric Interfaces

- Empowering Every Skill Level
- Accessibility through three distinct management tiers.
  - CLI (Command Line Interface): High-efficiency Makefile targets for power users and CI/CD pipelines.
  - Desktop GUI: A standalone PyQt5 application for point-and-click node management and real-time console logging.
  - Web UI: A Node.js-based interface for remote monitoring and management (in development).
  - Full Observability: Integrated logging of all operations for audit and troubleshooting.

Conclusion & Roadmap

- Efficiency, Scalability, and Sovereignty
- Autobuild delivers a professional virtualization experience using open-source tools.
  - Current State: Fully automated provisioning, role-based customization, and reliable backup/restore.
  - Key Benefits: Drastic reduction in deployment time, standardized node configurations, and ease of maintenance.
  - Future Outlook: Further expands of the Web UI, deeper integration with cluster management (GlusterFS), and support for additional distributions.
- “The platform that grows with your infrastructure”

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